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IoT based Smart Helmet for Bike Rider Safety

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***Abstract*— Numerous regulations have been implemented by the government to address the growing incidents of road accidents. Despite the establishment of comprehensive traffic regulations, the occurrence of accidents continues to rise. Adhering to safety measures like helmet usage and cautious driving can mitigate these issues. The adoption of helmets significantly reduces this risk. To enhance accident prevention and response, the integration of Internet of Things (IoT) technology has been introduced into smart helmets. This IoT-equipped helmet system serves multiple purposes, including accident prevention, prompt accident detection, and swift GPS-based position recovery. These functionalities collectively enhance the rider's confidence in their safety. The proposed strategy employs cost-effective sensors like NodeMCU and the MQ sensor. Notably, the helmet must be worn for the motorcycle to start, thanks to its ability to detect the rider's head within proximity. A Bluetooth module facilitates seamless wireless communication between the helmet and the motorcycle components. The most significant advantage of this approach lies in its incorporation of a fall detection mechanism utilizing the MPU6050 Sensor, coupled with a GPS module. This combination allows for rapid and accurate pinpointing of the accident's location, enabling swift response from emergency medical services. Simultaneously, pertinent information about the rider is relayed to emergency contacts. In essence, the smart helmet system greatly enhances accident detection, safety, and overall security for two-wheeled riders.**

Keywords*— *Arduino, Road accidents, Smart helmet, Global positioning system, Bluetooth, Traffic regulations.

I. INTRODUCTION

The advent of Internet of Things (IoT) technology [1] has brought about transformative advancements across various domains, and one such area is enhancing the safety of bike riders through the implementation of smart helmets [2-3]. These innovative helmets, integrated with IoT capabilities, offer a comprehensive solution to address the critical concerns surrounding bike rider safety. This introduction delves into the concept [4] and significance [5] of IoT-based smart helmets, shedding light on their potential to revolutionize the way we approach bike rider safety [6]. In recent years, the rise in road accidents involving bike riders has underscored the urgency of adopting proactive measures to mitigate risks and enhance overall safety (see Fig.1).

Conventional safety measures, such as traffic rules and regulations, have proven effective to a certain extent, yet the ever-evolving landscape of road safety demands more sophisticated and dynamic solutions [7-8]. This is where IoT-based smart helmets come into play, leveraging technology to create a safer environment for bike riders. At its core, an IoT-based smart helmet is not merely a protective headgear, but a cutting-edge device equipped with an array of sensors, connectivity modules, and data processing capabilities. This intricate combination enables the helmet to actively engage in accident prevention, timely accident detection, and efficient response mechanisms. By seamlessly integrating IoT elements into the helmet, a symbiotic relationship between technology and safety is forged.

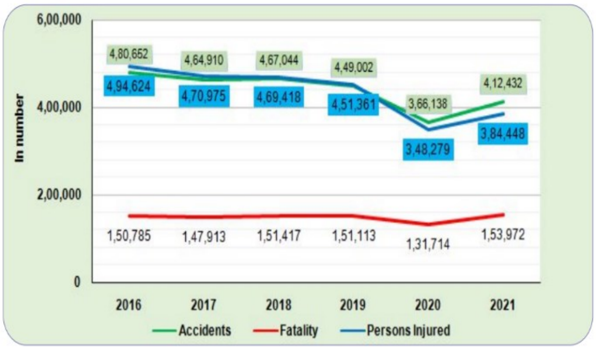


Fig. 1. Trends in accident occurrences, fatalities, and injuries sustained: from 2016 to 2021 [9].

Real-time accident detection utilizing cutting-edge sensor technology, prompt interaction with emergency services and designated contacts, and the ability to locate the exact location of the accident are some of the key characteristics of IoT-based smart helmets. These features collectively contribute to reducing response times, potentially saving lives, and ensuring prompt medical attention. Additionally, the helmets can provide valuable data insights, which can be utilized for accident analysis, infrastructure improvements, and proactive safety measures [10]. The purpose of this paper is to delve deeper into the mechanics and benefits of IoT-based smart helmets for bike riders. By exploring the technologies and components that make up these helmets, along with their integration into the broader IoT ecosystem, we aim to highlight the potential impact they can have on reducing accident rates, enhancing rider safety, and ushering in a new era of road safety strategies. The proposed work introduces the E-helmet, which prevents the rider from activating the bike's ignition unless they are wearing it. This E-helmet acts as a wireless key fob for a bicycle; without it, the bike would not start. In addition, the bike will not start if the rider is intoxicated, since an alcohol sensor will check his breath at every start.

II. LITERATURE REVIEW

Motorcycle riding is a popular mode of transportation that comes with inherent risks. To mitigate these risks and improve rider safety, researchers and developers have turned to IoT technologies to create smart helmets. These helmets integrate sensors, communication modules, and data processing capabilities to provide real-time monitoring and safety features for riders. Majority of the research has focused on accident detection and notification; alcohol detection; obstacle detection and warning; fall detection and emergency assistance; navigation and GPS integration.

The authors of [11] have developed a smart helmet that incorporate accelerometers and gyroscopes to detect accidents or collisions. When an accident is detected, the helmet can send an automatic distress signal to emergency contacts, helping ensure timely medical assistance. The authors in [12] have explored adding alcohol detection systems to smart helmets. These systems measure the rider's alcohol level through breath or sweat sensors, preventing the bike from starting if the alcohol level is above the legal limit. IoT-enabled helmets can utilize proximity sensors and cameras to detect obstacles and potential hazards on the road. When an obstacle is detected, the helmet can provide real-time warnings to the rider, as demonstrated in [13]. Smart helmets with fall detection capabilities, like the one proposed by [14], can sense sudden changes in acceleration and orientation that indicate a fall. These helmets can then send alerts to emergency contacts or authorities. GPS-enabled helmets can provide turn-by-turn navigation instructions to riders, displayed on a heads-up display. In [15], the authors have discussed the integration of GPS navigation in smart helmets for enhanced rider convenience and safety. Some intelligent helmets [16] have sensors to keep an eye on the rider's vital signals, such as heart rate and body temperature. This data can be sent to a mobile app for real-time health tracking, contributing to rider safety by identifying any health issues during the ride.

While IoT-based smart helmets offer promising advancements in rider safety, challenges remain. Ensuring the reliability of sensor data, optimizing power consumption, and addressing privacy concerns are important considerations for future research [17]. The integration of sensors, communication modules, and data analytics offers a range of features such as accident detection, alcohol sensing, obstacle warnings, and health monitoring. Despite challenges, these helmets hold immense potential to reduce accident-related injuries and fatalities, making motorcycle riding a safer mode of transport.

III. PROPOSED MODEL

A. Block Diagram

The first issue concentrates to prevent not wearing helmet and detect the rider is drunken drive these factors are the main reason for the accident cause. If the helmet is not wearied properly the ignition of the bike is blocked and will not start and when the helmet is wearied by the rider it will also check for alcohol condition if the rider consumed any alcohol, it will detect and the ignition will automatically stop for this module IR sensor and MQ sensors are used. The push switch is fitted inside the helmet so it can detect the rider head. This eliminates the case of the rider keep the helmet on the top of bike fuel tank. The MQ gas sensor is affixed to the helmet, positioned directly in front of the rider's mouth. It detects the alcohol and set the value to HIGH means the alcohol is detected and bike ignition will stop. The Bluetooth Module which is used for communication in between helmet unit as transmitter and bike unit as receiver. The final issue is when a rider met with an accident it takes more time to get medical help and rider's friend and family doesn't know about what has happened to him, to overcome this issue IoT and GPS modules [18] are used to detect the location where the accidents occur and send the location to the emergency user saved inside the device.

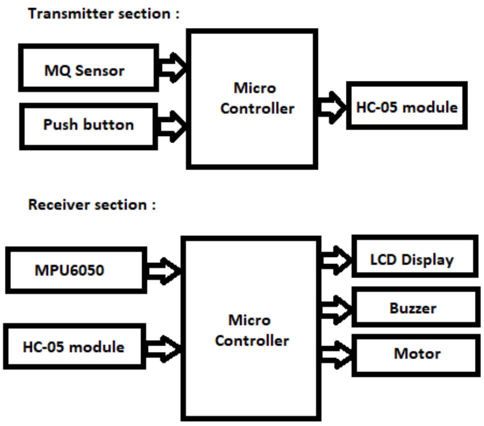
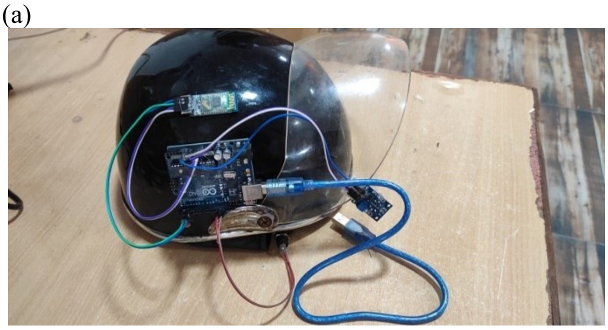


Fig. 2. Block diagram of proposed model.

B. Proposed Hardware Prototype

The proposed hardware prototype consists of various sensors interfaced with NodeMCU which is depicted using Fig. 3.

(a)



(b)

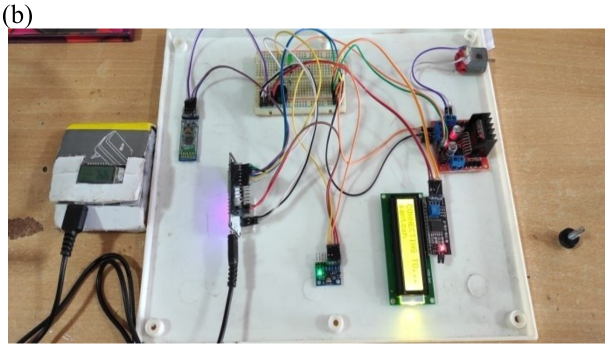


Fig. 3. Prposed Hardware Model (a) Transmitter Unit (b) Receiver Unit.

IV. PROCEDURE

A flow chart for the transmitter and receiver components of the Smart Helmet hardware design model has been built to explain how it functions for the safety of bike riders.

Bluetooth is used to transmit data from the bike's helmet to the bike's different operating stages so that each unit can be understood in detail.

- Step 1: Reset every piece of hardware.
- Step 2: When the helmet is turned ON it checks for receiver Bluetooth to connect.
- Step 3: . It checks the button status for helmet wearing information and checks alcohol data.
- Step 4: Button status and alcohol ranges are transmitted to receiver Bluetooth from the helmet section.

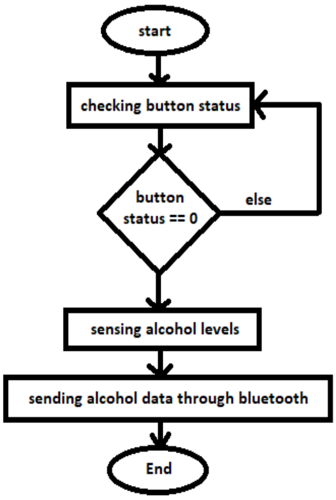


Fig. 4. Flow chart for transmitter section.

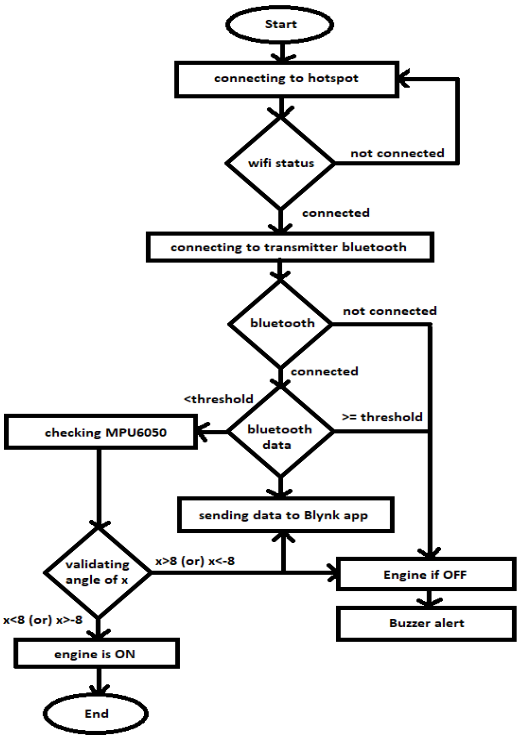


Fig. 5. Flow chart for receiver section.

- Step 5: At the bike section, initially it connects to hotspot network of registered SSID and Password, Bluetooth collects the data from the transmitter and MCU cheeks for the threshold levels of alcohol and button status to turn ON the Engine.
- Step 6: All the data is displayed by the mobile application which is connected to the MCU with the specified authentication key.

V. Experimental Results

The programming for the proposed system was performed using Arduino IDE (version R3, 5Watts, 7-12V), while Blink App has been utilized for creating the user interface. The push switch is fitted inside the helmet so it can detect the rider head. This eliminates the case of the rider keep the helmet on the top of bike fuel tank. The MQ gas sensor is fitted in front of the rider mouth of the helmet. It detects the alcohol and send the value to micro controller. All the information from the helmet is transmitted through Bluetooth which is connected in serial to the micro controller.

The experimental results demonstrates that the LCD display effectively showcases the sensor data obtained from helmet. the system Through the Bluetooth module various alters (wearing helmet (see Fig.6), alcohol detection when the rider alcohol intake level exceeded threshold (see Fig.7) and accident detection alert (see Fig.8)) have been sent to the LCD display.

Fig. 6. Helmet not wearied alert shown in bike display.

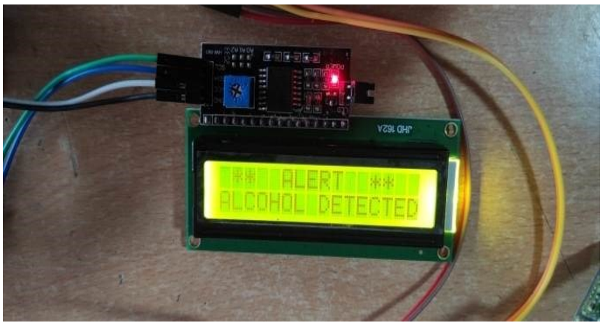


Fig. 7. Alcohol alert shown in bike display

Fig. 8. Accident alert shown in bike display

Bike unit displays the message like, “please wear the helmet”, “alcohol is detected”, “accident detected” and the levels are also displayed to know the present status. IoT which is used in receiver is used to share the all the information to the particular mobile application which is having an authentication key linked with MCU unit through the programming (or) instructions whitened by user.

Fig. 9. Status of mobile application (a) alchol level bellow the threshold (b) alchol level above the threshold (c) when qaccident caused.

VI. Conclusion

The primary objective of the proposed research work is to ensure the safety of riders by enforcing the mandatory use of protective gear and verifying the absence of alcohol consumption prior to or during the trip. When a rider violates any safety regulations, the ignition remains inactive, preventing the bike from starting. This approach effectively addresses various issues that afflicted earlier initiatives. The technology integrated into the helmet unit functions as a supportive feature when the bike attains its maximum speed, recognizing nearby vehicles and verbally alerting the rider. In case of an accident, the reporting system can promptly notify loved ones through alerts, providing accurate location details for immediate medical assistance. Moreover, this solution is cost-effective. A future enhancement could involve the ability to identify road potholes and speed bumps via an upgraded vehicle detection model. To optimize the microcontroller's efficiency, a specialized embedded system is being developed to process sensor data.

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